

PROFESSIONAL SUMMARY

Policy Researcher with expertise in program evaluation, causal inference, and applied machine learning, specializing in rigorous quasi-experimental and experimental methods, including propensity score matching, difference-in-differences, regression discontinuity, and instrumental variables. Experienced in designing and executing end-to-end research and evaluation projects using large-scale administrative, survey, and behavioral datasets. Proficient in R and Python. Proven ability to translate complex quantitative findings into clear, actionable insights for policymakers and business leaders.

EDUCATION

COLUMBIA UNIVERSITY, School of International and Public Affairs (SIPA) New York, USA
Master of Public Policy & Administration, Data Science for Policy (*Program Evaluation*) May 2026
Relevant Coursework: Causal Inference & Program Evaluation, Applied Econometrics, Time Series Analysis, Applied Machine Learning, Micro & Macro Economics

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS, Engineering College Dhahran, Saudi Arabia
Bachelor of Science, Electrical Engineering (*Signal Processing & Artificial Intelligence Minors*) September 2021

RESEARCH EXPERIENCE

INTERNATIONAL LABOUR ORGANIZATION (ILO) New York, USA
Graduate Researcher November 2025 - Present

- Conducted institutional and legal analysis of national forced labour frameworks, mapping enforcement structures and governance mechanisms to ground empirical findings in real-world policy and regulatory context.
- Translated the ILO's legal definition of forced labour into a structured measurement framework, aligning international standards with nationally representative microdata to produce a scalable foundation for empirical policy analysis.
- Designed and implemented econometric strategies, including propensity score matching and logistic regression, using nearly 70,000 individual-level survey observations to estimate associations between forced labour and socioeconomic vulnerability factors.
- Designed a victim typology framework using PAM clustering with Gower distance similarity and MDS embedding to segment forced labour-affected individuals into distinct profiles defined by demographic, economic, and behavioral characteristics, producing classification outputs to support evidence-based policy targeting.
- Built a random forest model to predict the likelihood of individuals being in forced labour based on victim profiles and forced labour indicators.
- Contribute to research briefs and technical documentation, translating statistical findings into policy-relevant insights.

PROFESSIONAL EXPERIENCE

COMMUNICATION, SPACE & TECHNOLOGY COMMISSION (CST) Riyadh, Saudi Arabia
Technology Strategist April 2023 - August 2024

- Designed and implemented a comprehensive assessment framework for 32 cloud service providers, and developed a performance index that informed \$30M in national investment policies, enhanced market transparency, and fostered competition by incentivizing providers to improve quality and efficiency.
- Built Tableau dashboards enabling real-time monitoring of provider performance and reducing reporting and decision-making time by approximately 60%.
- Produced analytical reports analyzing 1,200 organizations across 9 industries, identifying performance gaps and market demand trends to guide strategic growth and optimize investment decisions.
- Collaborated with policy, engineering, and investment teams to translate complex analyses into actionable strategies, driving 5 regulatory reforms in the national cloud regulatory framework.
- Produced executive policy briefs for senior government officials, including ministers, governors, and deputy governors, providing evidence-based assessments of the cloud services market, provider landscape, and emerging policy considerations to support strategic decision-making.
- Developed data-driven infographics and visual communication products for national awareness and policy campaigns, enhancing stakeholder engagement and improving the accessibility of complex technical and policy information.

DELOITTE Riyadh, Saudi Arabia
Business Analyst April 2022 - April 2023

- Assessed a financial-sector client's digital infrastructure, mapping services across 230 systems through surveys and interviews to identify gaps, mitigate risks, and address regulatory violations.

- Developed predictive models to estimate workload and resource demand, improving infrastructure planning accuracy and reducing potential system bottlenecks.
- Designed a hosting and data management strategy ensuring regulatory compliance and customer data protection while optimizing system load, improving overall system reliability by 40% and critical system reliability by 100%.
- Conducted a current-state assessment of a government agency, mapping services and digital infrastructure across 11 sub-agencies, identifying inefficiencies and driving the integration of 66 systems, reducing operational costs by 20% and improving infrastructure investment efficiency.
- Presented analytical findings and risk evaluations to senior leadership, shaping digital transformation decisions and informing long-term implementation roadmaps.

SAUDI TELECOM COMPANY (STC)

Riyadh, Saudi Arabia

Access & Transport Strategy Developer

December 2021 - April 2022

- Applied predictive and geospatial analysis to inform infrastructure investment decisions aligned with national connectivity and inclusion goals.
- Assessed regional disparities in service access to support equitable expansion planning.
- Developed a data-driven insights report to support strategic planning and align infrastructure decisions with national objectives.

ARAMCO

Ras Tanura, Saudi Arabia

Control and Network Engineer

September 2021 - December 2021

- Supported the migration and modernization of mission-critical industrial control systems underpinning large-scale energy export operations, including SCADA, PLC, and DCS architectures.
- Coordinated with cross-functional engineering and safety teams to align control systems with reliability, cybersecurity, and operational governance standards.

LEADERSHIP & ACTIVITIES

Program Assistant – Data Science for Policy Concentration at SIPA

Sep 2025 - May 2026

- Coordinate academic programming, workshops, and events bridging data science and public policy.
- Collaborate with faculty, administrators, and students to improve program operations and enhance student engagement.
- Support curriculum development and resources to strengthen applied data science training for policy students.
- Led hands-on workshops mentoring students in Python and R programming, data manipulation, and applied analytics.

CERTIFICATIONS

- **Advanced Regulations** - University of Oxford, Oriel College
Completed an intensive program on advanced regulatory frameworks, including antitrust, competition law, and governance mechanisms. Developed expertise in navigating complex regulatory environments, enhancing strategic decision-making, and applying theoretical knowledge to practical regulatory challenges.
- **Digital Regulations** - International Telecommunication Union (ITU)
Completed a professional program on digital regulation, covering regulatory governance, competition, consumer protection, data privacy, spectrum management, emerging technologies, two-sided platforms, and digital inclusion. Gained practical skills in applying regulatory frameworks, developing evidence-based policies, and navigating the complexities of the digital economy.

SKILLS

- **Statistical & Econometric Methods:** OLS and generalized linear regression, logistic regression, fixed effects, difference-in-differences, instrumental variables (2SLS), regression discontinuity, propensity score matching, time series modeling and ARIMA-based forecasting, and GARCH and VAR models.
- **Machine Learning:** Decision Trees, Random Forest, LASSO/Ridge Regularization, k-Nearest Neighbor, PAM, and K-Modes Clustering
- **Programming & Tools:** R (tidyverse, ggplot2, MatchIt, plm, ivreg, knitr, survey), Python (pandas, sklearn, statsmodels), Stata, SQL, GitHub, MS Office Suite (advanced).
- **Research & Communication:** Academic research writing, client-facing report preparation, codebook documentation, and data visualization.